Power BI Project Documentation

Blinkit Analytics Dashboard

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1. Project Objective

The primary objective of this project is to:

Analyze sales performance across products and outlets

Understand customer satisfaction using ratings

Evaluate product visibility, weight, and fat content impact on sales

Provide data-driven insights to improve:

Product strategy

Outlet performance

Inventory distribution

2. Project Overview

This project is built using Power BI and focuses on analyzing Blinkit's retail dataset.

It is divided into two dashboards:

Dashboard Page

Focus Area

Page 1

Sales & Outlet Performance

Page 2

Customer & Product Insights

The analysis leverages derived columns and DAX measures to uncover meaningful insights.

3. Dataset Description

The dataset contains the following key fields:

Item Fat Content

Item Identifier

Item Type

Outlet Establishment Year

Outlet Identifier

Outlet Location Type

Outlet Size

Outlet Type

Item Visibility

Item Weight

Sales

Rating

4. Derived Columns

Derived Column	DAX	Description
Outlet Age	`Outlet age = YEAR(TODAY())-'BlinkIT Grocery Data'[Outlet Establishment Year]`	Difference between current year and outlet establishment year
Sales Bucket	`Sales Bucket = SWITCH(TRUE(),'BlinkIT Grocery Data'[Sales] < 100, "Low", 'BlinkIT Grocery Data'[Sales] <= 200, "Medium","High")`	Categorizes sales into Low, Medium, High
Weight Group	`Weight Group = IF('BlinkIT Grocery Data'[Item Weight] < 10, "Light","Heavy")`	Categorizes items into Light and Heavy
Visibility Category	`Visibility Category = IF('BlinkIT Grocery Data'[Item Visibility] = 0,"Not visble", IF('BlinkIT Grocery Data'[Item Visibility] < 0.05, "Low","High"))`	Categorizes visibility into High, Low, Not Visible

5. DAX Measures

Measure Name	DAX	Description
Total Sales	`Total sales = SUM('BlinkIT Grocery Data'[Sales])`	Sum of all sales
Total Items Sold	`Total Items sold = COUNT('BlinkIT Grocery Data'[Item Identifier])`	Count of item identifiers
Top Item Sales	`Top Item Sales = MAXX(VALUES('BlinkIT Grocery Data'[Item Type]), [Total sales])`	Highest sales among item types
Sales per Outlet	`Sales Per Outlet = DIVIDE([Total sales], DISTINCTCOUNT('BlinkIT Grocery Data'[Outlet Identifier]))`	Average sales per outlet
High Visibility Sales	`High Visibility Sales = CALCULATE([Total sales], 'BlinkIT Grocery Data'[Visibility Category] = "High")`	Sales for high visibility items
Low Visibility Sales	`Low Visibility Sales = CALCULATE([Total sales], 'BlinkIT Grocery Data'[Visibility Category] = "Low")`	Sales for low visibility items
Not Visibility Sales	`Not Visibility Sales = CALCULATE([Total sales], 'BlinkIT Grocery Data'[Visibility Category] = "Not visible")`	Sales where visibility is zero

Average Sales	`Avg Sales = AVERAGE('BlinkIT Grocery Data'[Sales])`	Average sales per item
Average Rating	`Average Rating = AVERAGE('BlinkIT Grocery Data'[Rating])`	Average customer rating

6. Dashboard – Sales & Outlet Performance

Purpose:

To analyze revenue trends and outlet performance

Visuals Used

Bar Chart : Sales by Item Type & Fat Content

X-axis: Item Type

Y-axis: Total Sales

Legend: Item Fat Content

Insight: Identifies which product categories and fat types drive revenue

Column Chart: Sales by Outlet Location & Size

X-axis: Outlet Location Type

Y-axis: Total Sales

Legend: Outlet Size

Insight: Compares performance across different outlet types and sizes

Donut Chart: Sales by Fat Content

Legend: Item Fat Content

Values: Total Sales

Insight: Shows contribution of low-fat vs regular products

Line Chart: Sales Trend Over Time

X-axis: Outlet Establishment Year

Y-axis: Total Sales

Insight: Tracks sales growth over years

7. Dashboard – Page 2: Customer & Product Insights

Purpose:

To analyze customer behavior and product performance

Visuals Used

Column Chart: Sales by Visibility & Weight

X-axis: Visibility Category

Y-axis: Total Sales

Legend: Weight Group

Insight: Analyzes impact of product visibility and weight on sales

Scatter Plot: Rating vs Sales

X-axis: Average Rating

Y-axis: Total Sales

Legend: Item Type

Insight: Shows relationship between customer satisfaction and sales

Waterfall Chart: Sales Contribution Analysis

Category: Item Type

Breakdown: Item Fat Content

Values: Total Sales

Insight: Visualizes increase/decrease in sales contribution across categories

8. Key Project Highlights

Built a 2-page interactive dashboard

Used data transformation and derived columns

Created business-relevant KPIs

Performed multi-dimensional analysis:

Product

Customer

Outlet

Applied visual storytelling techniques

9. Key Business Insights

High visibility items contribute significantly to sales

Certain item types dominate revenue generation

Customer ratings show a moderate correlation with sales

Outlet size and location influence performance

Low-fat products show strong contribution in some categories

10. Final Conclusion

This Blinkit Analytics Dashboard provides a comprehensive view of retail performance by combining:

Sales analysis

Customer behavior insights

Product-level evaluation

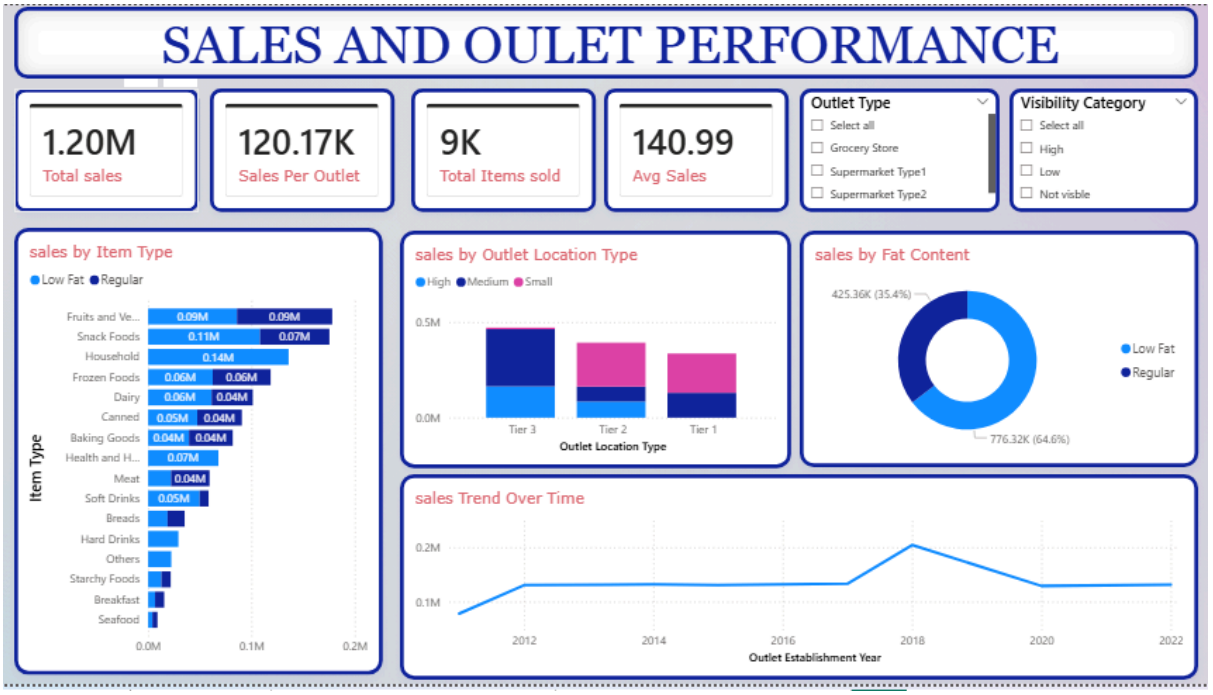
The dashboard enables stakeholders to:

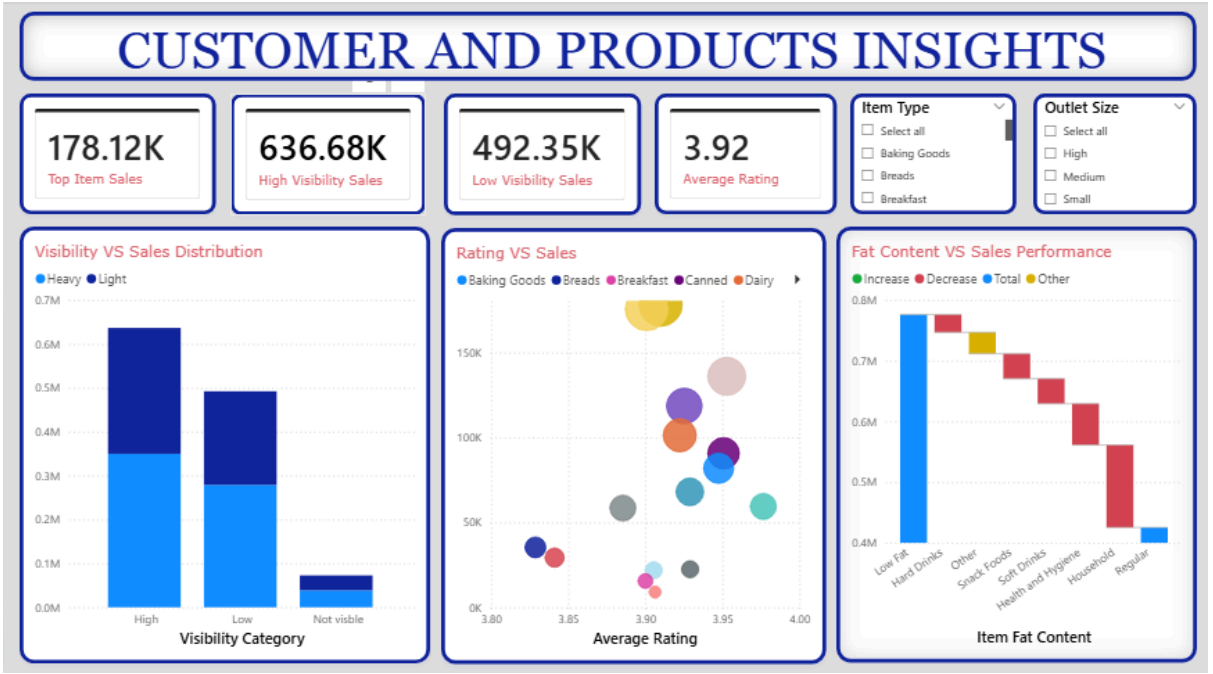
Make data-driven decisions

Improve inventory and product strategy

Enhance customer satisfaction

Optimize outlet performance


SALES AND OULET PERFORMANCE
1.20M
Total sales
120.17K
Sales Per Outlet
9K
Total Items sold
140.99
Avg Sales
Outlet Type
Select all
Grocery Store
Supermarket Type1
Supermarket Type2
Visibility Category
Select all
High
Low
Not visble
sales by Item Type
Low Fat
Regular
Item Type
Fruits and Ve...
Snack Foods
Household
Frozen Foods
Dairy
Canned
Baking Goods
Health and H...
Meat
Soft Drinks
Breads
Hard Drinks
Others
Starchy Foods
Breakfast
Seafood
0.0M
0.1M
0.2M
sales by Outlet Location Type
High
Medium
Small
0.5M
0.0M
Tier 3
Tier 2
Tier 1
Outlet Location Type
sales by Fat Content
425.36K (35.4%)
776.32K (64.6%)
Low Fat
Regular
sales Trend Over Time
0.2M
0.1M
2012
2014
2016
2018
2020
2022
Outlet Establishment Year


CUSTOMER AND PRODUCTS INSIGHTS
178.12K
Top Item Sales
636.68K
High Visibility Sales
492.35K
Low Visibility Sales
3.92
Average Rating
Item Type
Select all
Baking Goods
Breads
Breakfast
Outlet Size
Select all
High
Medium
Small
Visibility VS Sales Distribution
Heavy
Light
0.7M
0.6M
0.5M
0.4M
0.3M
0.2M
0.1M
0.0M
High
Low
Not visble
Visibility Category
Rating VS Sales
Baking Goods
Breads
Breakfast
Canned
Dairy
150K
100K
50K
0K
3.80
3.85
3.90
3.95
4.00
Average Rating
Fat Content VS Sales Performance
Increase
Decrease
Total
Other
0.8M
0.7M
0.6M
0.5M
0.4M
Low Fat
Hard Drinks
Other
Snack Foods
Soft Drinks
Health and Hygiene
Household
Regular
Item Fat Content